

Force-free molecular dynamics through autoregressive equivariant networks

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Molecular dynamics simulations play a crucial role in scientific research. Yet their computational cost often limits the timescales and system sizes that can be explored. Most data-driven efforts have focused on reducing the computational cost of the accurate interatomic forces required to solve the equations of motion. However, despite their success, these machine-learning interatomic potentials are still bound to small time steps. In this work, we introduce TrajCast, a transferable and data-efficient framework based on autoregressive equivariant message-passing networks, that directly updates the atomic positions and velocities, lifting the constraints imposed by traditional numerical integration. We benchmark our framework across various systems, including a small molecule, crystalline material and bulk liquid, demonstrating excellent agreement with reference molecular dynamics simulations for structural, dynamical and energetic properties. Moreover, we show that TrajCast can generalize in a zero-shot manner to unseen regions of phase space, producing physically meaningful ensembles in metastable equilibrium and out-of-equilibrium regimes beyond the training data, without compromising accuracy. Depending on the system, TrajCast allows for forecast intervals up to 30× larger than traditional MD time steps, generating over 15 ns of trajectory data per day for a solid with more than 4,000 atoms. By enabling efficient, large-scale simulations over extended timescales, TrajCast can accelerate materials discovery and explore physical phenomena beyond the reach of traditional simulations and experiments.

Molecular dynamics (MD) simulations are an essential tool in modern science, providing mechanistic insight into complex phenomena such as protein folding^{1,2}, the effects of pressure on materials^{3–6}, the existence of liquid–liquid transitions in supercooled liquids^{7–9} and amorphization^{10–12}, to name but a few, that are difficult or impossible to study through experiments alone. By numerically integrating the equations of motion, these simulations generate a trajectory of atomic positions $\mathbf{r}(t)$ and velocities $\mathbf{v}(t)$ for all N atoms over time. However, obtaining trajectories at first-principles accuracy is computationally

intensive, typically limiting them to systems of a few hundred atoms and timescales up to 1 ns. These computational costs arise from (1) accurately approximating the Schrödinger equation to obtain reliable forces and (2) the necessity for millions of integration steps owing to the typically small time step of 0.5–1.0 fs imposed by the cubic scaling of the discretization error with the integration time step. Accordingly, two strategies have emerged to speed up trajectory generation, focusing on either improving the force prediction efficiency or evolving the system with larger time steps.

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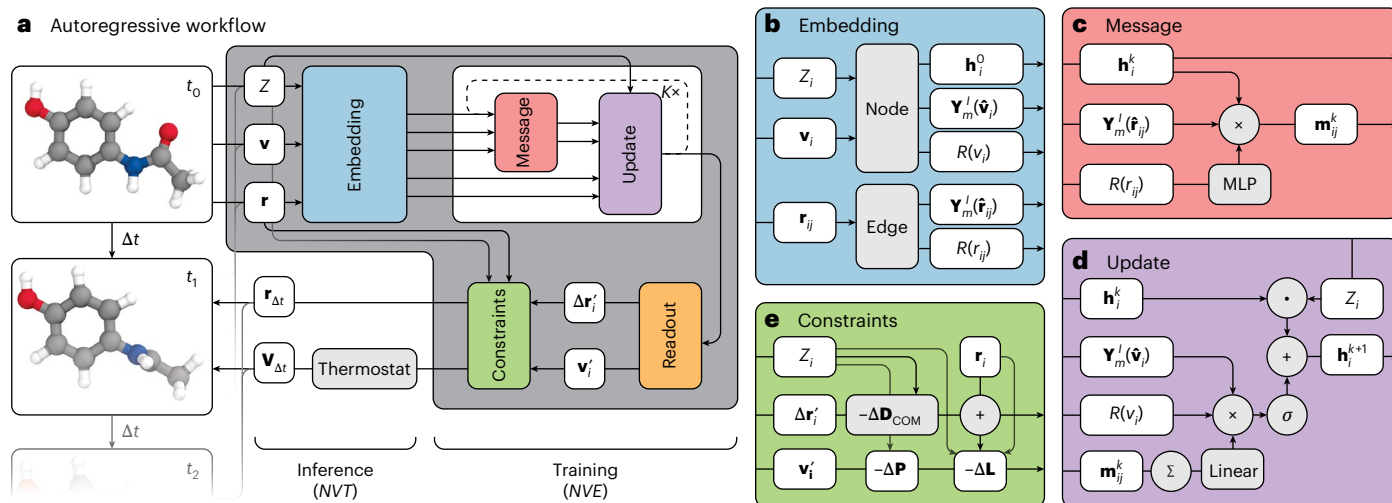


Fig. 1 | An overview of the TrajCast framework. a, The autoregressive workflow: An atomistic system at time t_0 is passed through an equivariant MPNN (grey box) to predict the new positions and velocities at time t_1 . Atomic attributes (positions, velocities and chemical elements) are encoded into the initial features, which are refined over T message-passing blocks. Estimates of the displacement and velocity vectors are generated on the basis of the final features and then refined to ensure momentum conservation. The weights of the MPNN are optimized by training on *NVE* trajectories. The trajectory is built by rolling out predictions, where outputs from one step serve as inputs for the next. During inference, a thermostat ensures sampling from the *NVT* ensemble at constant temperature T , with states following the Boltzmann distribution. In this atomistic system,

hydrogen atoms are shown in white, oxygen in red, carbon in grey, and nitrogen in blue. **b**, The embedding block encodes node and edge attributes and generates the initial features. **c**, Messages are constructed by convolving latent features with filters derived from a learnable radial basis and the spherical harmonics expansion of edge vectors. **d**, In the update block, messages from neighbours are pooled and combined via a tensor product with velocity vectors in a learnable radial and spherical harmonic basis. The result is passed through a nonlinearity and added to the previous layer's features, weighted by the node's chemical element. **e**, Conservation of total linear and angular momentum is enforced by adjusting the displacements and velocities.

Machine-learning interatomic potentials (MLIPs) fall into the first group and can accelerate the calculation of ab initio-quality forces by orders of magnitude^{13–16}. Trained on small systems for which electronic structure calculations are feasible, these models operate locally, allowing for an accurate representation of the potential energy surface in broader, chemically similar systems without requiring direct reference data. This has enabled access to much larger time and length scales, as demonstrated by the remarkable nanosecond-long simulation of the human immunodeficiency virus with 44 million atoms¹⁷ and has catalysed exciting discoveries in simulations, including a superionic phase of water under confinement¹⁸ and structural and electronic phase transitions in silicon¹⁹. The field has rapidly evolved from kernel-based methods to equivariant graph neural networks (GNNs)^{20–31}, culminating recently in foundation models trained on large parts of the Periodic Table^{32–36}, which can be fine-tuned to the system of interest.

Multiple time-step (MTS) integrators^{37,38}, conversely, follow the second strategy by assigning shorter time steps to rapidly changing interactions, such as atomic bonds, and longer time steps to slower ones. While commonly used with a classical potential owing their inherent scale separation, a recent study³⁹ successfully coupled MTS with MLIPs by co-training a smaller model for short-time interactions alongside a larger model called less frequently, achieving a $3\times$ speed-up. However, MTS integrators are fundamentally limited by the need for small time steps in parts of the system, and further validation of scale-separated MLIPs for more complex systems is needed. The recent surrogate BoostMD⁴⁰ uses node embeddings from a reference MLIP at previous time steps to predict energy and forces for subsequent configurations, allowing a faster model to run between less frequent, full MLIP evaluations.

Generative machine-learning approaches go beyond the idea of MTS, seeking to efficiently generate representative configurations or even complete trajectories without computing or integrating interatomic forces. Boltzmann generators⁴¹ directly approximate the Boltzmann distribution but produce temporally unordered samples, precluding dynamical insights. Traditional and recurrent neural

networks have been applied to interpolate MD trajectories⁴² and predict dynamics⁴³ with time steps over $1,000\times$ larger than in traditional MD simulations⁴⁴. However, these methods operate with a fixed particle number, restricting their applicability to large-scale problems where accurate ab initio reference data are often hard to obtain.

The flow-based Timewarp⁴⁵ model, conversely, generates MD trajectories via autoregressive roll-out with time steps up to the order of 10^5 fs and has been shown to generalize to unseen peptides. Notwithstanding the speed-up of Timewarp and its recent generative competitors ITO⁴⁶, MDGen⁴⁷ and LiFlow⁴⁸, these approaches propagate atomic positions only, leaving the system's state incomplete, without momenta. Beyond impeding the computation of transport properties, this lack of full phase-space coverage introduces ambiguity as the same configuration can correspond to different temperatures, limiting interpolation across thermodynamic variables. Other deep learning architectures, such as the equivariant graph neural operator⁴⁹ and the works presented by Fu et al.⁵⁰ and Nam et al.⁴⁸, incorporate assumptions about chemical bonds or specific atomic motion patterns, or restrict their focus to heavy atom motion. While these inductive biases may improve the performance on specific tasks, they hinder generalizability to reactive systems in which bond breaking and formation occurs. Moreover, most generative models require hundreds of nanoseconds of training data and have been evaluated exclusively on small molecules and peptides, leaving their applicability to materials and condensed matter largely unexplored.

In this work, we introduce TrajCast, a framework that employs equivariant message-passing neural networks (MPNNs) to autoregressively predict the next state of a molecular system. Unlike most MLIPs that derive forces from predicted energies, TrajCast directly outputs the next state (positions and velocities) over a period Δt that is at least one order of magnitude larger than the typical time step in equivalent MD simulations. By preserving the sequential nature of the trajectory, TrajCast recovers the system's dynamics and enables the computation of time-dependent properties. Generating full-state trajectories ensures a complete system representation, enabling interpolation and

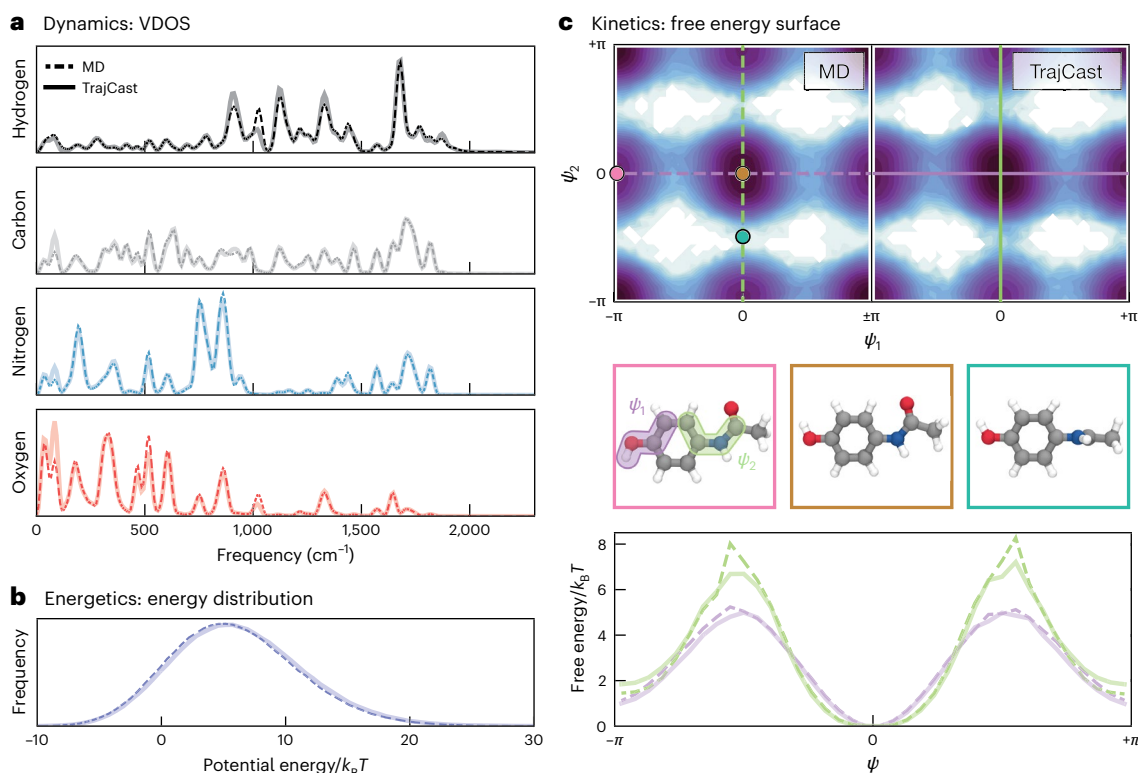


Fig. 2 | A comparison of the properties of paracetamol from MD and TrajCast with a 7-fs prediction horizon. a, Dynamical properties are compared on the basis of the element-specific VDOS. **b**, Energetics are validated by comparing the distribution of the potential energy, computed in post-processing for the TrajCast-generated trajectory by using the same force field as in the reference MD. **c**, Free energy barriers drive the system's kinetics and are computed from the

FES, shown for both MD and TrajCast at the top. The FES is generated by sampling two dihedral angles, ψ_1 and ψ_2 , visualized in the centre panel, where hydrogen is shown in white, oxygen in red, carbon in grey, and nitrogen in blue. The centre panel visualizes different realizations of paracetamol for various dihedral pairs, matching the colours of corresponding points in the FES. Barriers are calculated along the cuts shown on the FES and coloured accordingly in the bottom panel.

extrapolation across different temperature regimes. TrajCast is free of assumptions about chemical bonds, making it capable, in principle, of modelling reactive systems. Relying on local environments, TrajCast is highly scalable and transferable, enabling models trained on small systems to be applied to larger or chemically similar ones. Moreover, TrajCast is very data-efficient, requiring far less than 1 ns of trajectory training data to achieve accurate predictions. We demonstrate its accuracy across a range of systems, from small molecules to crystalline solids, bulk liquids and out-of-equilibrium, kinetically arrested dynamics, highlighting its ability to consistently reproduce structural, energetic and dynamic properties and showcasing its robustness and versatility across different domains. Additionally, we show that TrajCast can generalize beyond its training distribution and explore unseen regions of phase space. Dependent on the system, TrajCast uses time intervals between $10\times$ and $30\times$ larger than traditional MD time steps. By avoiding gradient computations, this enables trajectory generation of more than 15 ns and 1 ns per day for crystalline quartz and liquid water, comprising 4,300 and 5,100 atoms, respectively.

Results

TrajCast framework

We start by outlining our framework to predict the temporal evolution of an atomistic system, as depicted in Fig. 1. TrajCast produces trajectories autoregressively, where the state $\mathbf{x}(t + \Delta t)$ is generated from $\mathbf{x}(t)$ and used to predict the next state at the subsequent time interval Δt (Fig. 1a), thus preserving the Markovian nature of traditional MD. Central to our proposed methodology is an equivariant MPNN that takes atomic positions $\mathbf{r}(t)$, velocities $\mathbf{v}(t)$ and atomic types Z as input. Rather than estimating the atomic forces and using numerical integration with a small time step δt , our MPNN directly predicts the new positions

$\mathbf{r}_{\Delta t} = \mathbf{r}(t + \Delta t)$ and velocities $\mathbf{v}_{\Delta t} = \mathbf{v}(t + \Delta t)$ at a larger time interval $\Delta t \gg \delta t$, as shown in the grey box in Fig. 1a. Specifically, the MPNN approximates the solution of Newton's equations of motion over Δt , evolving the system within the microcanonical number of particles (N), volume (V), total energy (E) (NVE) ensemble. To generate trajectories from the canonical N , V , temperature (T) (NVT) ensemble and enable comparison with experiments typically performed at constant temperature, we apply the canonical sampling through velocity rescaling (CSVR) thermostat⁵¹ after each MPNN forward pass and rescale the predicted velocities to enforce the desired temperature. Additional background on equivariant MPNNs, statistical ensembles and full architectural and training details are provided in the Methods.

Experiments

We validate the accuracy of TrajCast across three systems: a gas-phase paracetamol molecule, crystalline α -quartz and liquid water, training several models at different prediction horizons Δt . For each system, we present the largest Δt that produces stable trajectories and report mean absolute errors (MAE) in Supplementary Section A.1. Since the CSVR thermostat introduces stochasticity, direct trajectory comparison is infeasible, and instead, as is common practice, we compare ensemble averages of dynamic, structural and energetic properties. We show in Supplementary Section B.2 that TrajCast can closely follow MD NVE trajectories without a thermostat for short durations ($\sim 1,000$ steps), but the thermostat is necessary for nanosecond-scale simulations. Finally, we compare the data efficiency across systems and prediction horizons.

Small molecule: paracetamol. We first benchmark TrajCast on paracetamol, a small molecule with four distinct chemical elements (H, C, N and O) included in the (rev)MD17 datasets^{52,53}. We train on

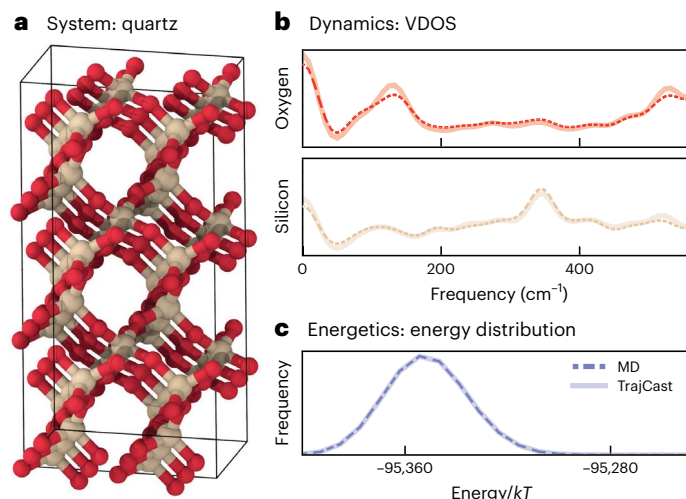


Fig. 3 | An evaluation of TrajCast on crystalline quartz with a 30-fs prediction horizon. **a**, An illustration of an atomic configuration of quartz used to train and validate TrajCast. Red spheres represent oxygen atoms, while brown spheres represent silicon atoms. **b**, Dynamical properties compared on the basis of the element-specific VDOS. **c**, Energetics validated by comparing the distribution of the potential energy, computed in post-processing for the TrajCast-generated trajectory by using the same force field as in the reference MD.

10,000 *NVE* configurations at 300 K and set the prediction horizon to $\Delta t = 7$ fs, $14\times$ the MD time step. For all the systems studied in this work, the simulation length is chosen to converge the properties examined, rather than being imposed by methodological limitations, and could in principle be extended. With the CSVR thermostat, we generate a stable 7-ns trajectory (10^6 steps) and validate the dynamics, energetics and kinetics with respect to MD *NVT* trajectories (Fig. 2). All the properties from MD were computed using the same frame frequency as the prediction horizon in TrajCast.

The species-resolved vibrational density of states (VDOS) in Fig. 2a shows excellent agreement with MD within the resolvable frequency range ($\sim 2,400$ cm^{-1} for $\Delta t = 7$ fs), achieving overlap scores⁵⁴ of ~ 0.95 . Unlike larger MD time steps, TrajCast does not introduce peak shifts from integration artefacts. The potential energy distribution (Fig. 2b), computed in post-processing with the same force field, matches closely, confirming that TrajCast samples from the Boltzmann distribution, indicating that structural features of the molecule are well recovered.

We further assess TrajCast's kinetics by computing the free energy surface (FES) as a function of two dihedral angles, ψ_1 and ψ_2 , corresponding to the hydroxyl and amine torsional rotations (Fig. 2c). The FES from MD and TrajCast agree well, with both capturing the *cis* and *trans* configurations of paracetamol. Quantitatively, the barrier profiles along cuts at $\psi_1 = 0$ and $\psi_2 = 0$ closely match, with deviations within $\sim k_B T$, where k_B is Boltzmann's constant. The amine flip ($|\psi_2| > \pi/2$) is separated by a barrier of $\gtrsim 7k_B T$, defining distinct metastable basins. While this conformational transition can be captured with long unbiased simulations, it is noteworthy that the training set contains no configurations with $|\psi_2| > \pi/2$ (Supplementary Section C.1). Despite this, TrajCast correctly generates the flipped state during roll-out and recovers the corresponding free energy barrier, demonstrating its ability to escape known minima and generalize beyond its training distribution.

Overall, the agreement in kinetics, energetics and dynamics provides strong evidence that TrajCast reliably predicts the trajectories of isolated molecules, implicitly encoding the potential energy surface by learning the conditional probability of local atomic motions.

Crystalline solid: quartz. We now move to benchmarking our framework for condensed matter phases, beginning with crystalline α -quartz, illustrated in Fig. 3a. To this end, we train our MPNN on 5,000 samples

from MD simulations of systems comprising 162 atoms at 300 K. In contrast to gas-phase molecules, particles forming a periodic lattice exhibit a more constrained and correlated motion, which allows us to set the prediction horizon to $\Delta t = 30$ fs, a 30 fold increase over the MD time step, while still generating stable and accurate trajectories. Similar to the previous section, we compare TrajCast and MD on the basis of the VDOS and energy distribution, as shown in Fig. 3b and 3c, respectively. These properties are calculated from trajectories of 15 ns length corresponding to 500,000 iterative TrajCast predictions.

We find that TrajCast agrees very well with the MD reference for both quantities, achieving VDOS overlap scores⁵⁴ above 0.95 for both atom types within the resolvable frequency range and an almost identical energy distribution. The peak at zero frequency is an artefact caused by the small system size and low time-sampling resolution. We observe that it disappears in both MD and TrajCast when the trajectory is sampled at intervals of 10 fs or lower. The accurate reproduction of the dynamics and energetics at large time intervals suggests that TrajCast is a useful tool for studying the equilibrium properties of solids. In Supplementary Section B.3, we demonstrate that our approach scales efficiently with system size, enabling the generation of nearly 20 ns per day for a $3\times 3\times 3$ quartz supercell comprising about 4,300 atoms. As a comparison, MD simulations using the widely used MLIP MACE architecture typically produce around 1 ns of trajectory per day for 1,000 atoms³⁶. Beyond pristine lattices, this could make TrajCast well-suited for exploring defective materials, where studying phenomena at low defect concentrations requires particularly large system sizes.

Liquid: bulk water. Finally, we train a TrajCast model for liquid bulk water (Fig. 4a) using 64 molecules at ambient conditions, a common setup in *ab initio* studies of bulk water⁵⁵, with 5,000 training samples and $\Delta t = 5$ fs. The prediction horizon is ten times the MD time step but smaller than that for quartz, reflecting the more complex dynamics of a molecular liquid. We generate a 2.5-ns trajectory (500,000 steps) for comparison with MD and validate TrajCast on the basis of the VDOS, mean squared displacement (MSD) and radial distribution function (RDF).

The VDOS (Fig. 4b) shows near-perfect agreement for both oxygen and hydrogen across the entire frequency range, with overlap scores⁵⁴ of ~ 0.99 . This strongly suggests that TrajCast accurately captures molecular motion across different modes, from high-frequency intramolecular bending and stretching to low-frequency intermolecular translation and libration. Complementing this picture, the MSD (Fig. 4c) agrees well in the ballistic regime ($\tau \lesssim 0.5$ ps), with slight divergence at longer timescales probably due to error accumulation. The diffusion coefficients of 2.18 ± 0.19 and $1.84 \pm 0.03 \times 10^{-9}$ $\text{m}^2 \text{s}^{-1}$ obtained using TrajCast and MD, respectively, confirm the reliable prediction of liquid dynamics. Beyond the dynamics, the RDF (Fig. 4d) shows excellent agreement across all pairwise element combinations, with only a slight understructuring in the O–O RDF, consistent with the marginally higher diffusion coefficient.

These benchmarks suggest that our methodology can accurately capture the dynamical and structural characteristics of bulk liquids at room temperature in an efficient manner. Even with $\Delta t = 5.0$ fs, TrajCast generates over 1 ns per day for a $3\times 3\times 3$ supercell with more than 5,000 atoms (Supplementary Section B.3).

Zero-shot generalization: hyperquenched glassy water. The preceding experiments demonstrated that TrajCast accurately reproduces properties of systems on which it was trained. However, many problems require simulations to explore unseen regions of phase space, such as the formation of supercooled and glassy states^{56–58}, posing a challenge for generative models to generalize beyond their training distribution. To test this, we performed a zero-shot evaluation using the water model trained exclusively at ambient conditions to predict the amorphization and formation of hyperquenched glassy water (HWG)^{59–64} upon cooling.

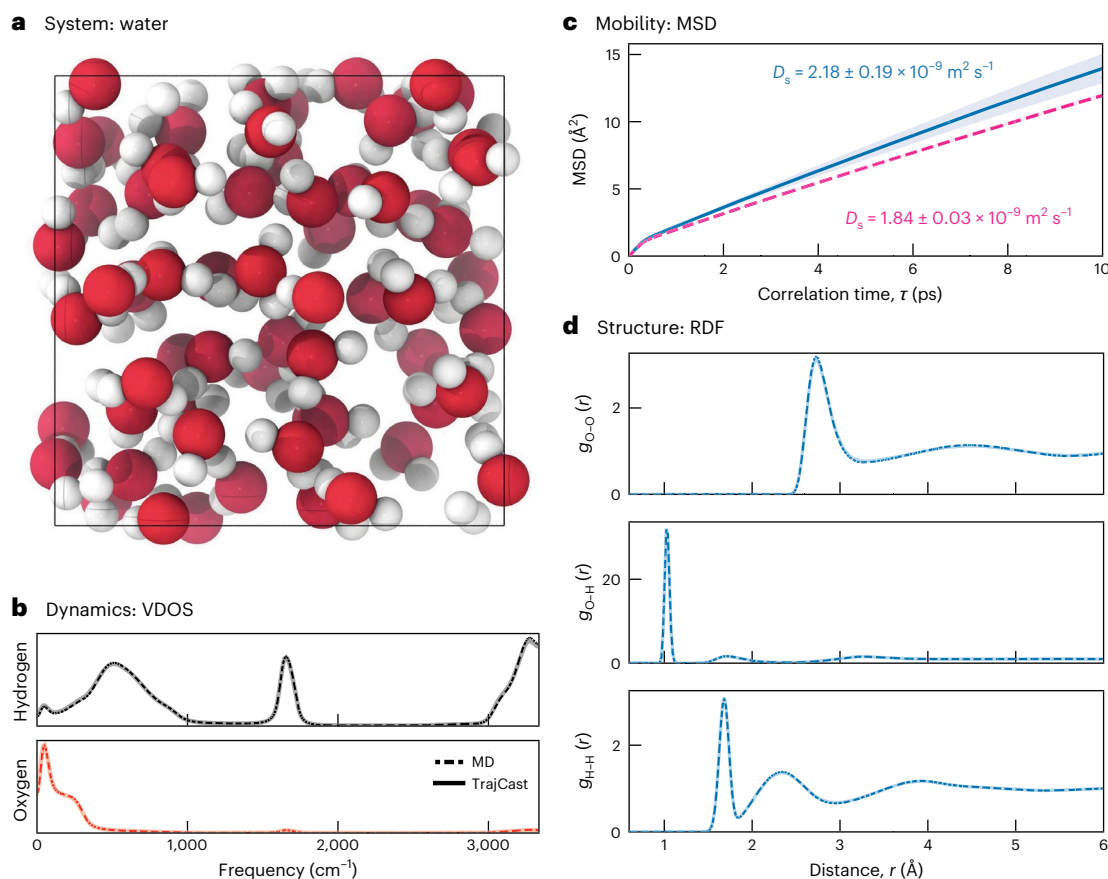


Fig. 4 | Benchmarking TrajCast on liquid water with a 5-fs prediction horizon. **a**, A snapshot of an instantaneous configuration of bulk water, with hydrogen atoms shown in white and oxygen atoms in red. **b**, A comparison of the element-specific VDOS between TrajCast and MD. **c**, The MSD of oxygen atoms over a correlation time τ of up to 10 ps. The diffusion coefficients are computed on

the basis of a fit to the slope in the range from 1 to 10 ps following the procedure outlined in ref. 55. The shaded areas represent the threefold standard deviation around the mean value, computed over four blocks. **d**, Validation of the molecular liquid structure on the basis of the pairwise combinations of chemical elements in the RDF.

HGW exhibits a complex potential energy landscape where different cooling protocols lead to distinct structural properties^{65–67}, making this a particularly challenging test. We performed four independent *NVT* roll-outs on a system of 216 water molecules, cooling from 300 to 180 K at a constant rate of $q = 10$ K ns^{−1}, consistent with a recent MLIP study⁶⁸, culminating in 12 ns and 2.4 million autoregressive steps per run.

Remarkably, all four trajectories remain stable throughout the cooling process, despite being driven far beyond the temperature range seen during training. Figure 5 compares the dynamics and structure of water with reference MD simulations (for comprehensive results, see Supplementary Section C.2). The diffusion coefficient (panel a) closely matches MD across the entire temperature range, with a small offset that vanishes at $T \approx 210$ K, coinciding with the onset of dynamical arrest where the system enters a glassy state. The oxygen–oxygen RDFs at selected temperatures (panel b) are nearly indistinguishable between TrajCast and MD. Upon cooling, the first and second RDF peaks sharpen, while the inter-peak region becomes progressively depleted of water molecules, consistent with literature reports for HGW⁶⁵.

Following the analysis of ref. 65, we assessed the structural dynamics by using the order parameters τ and Q , which quantify the long-range translational and short-range orientational order, respectively (Methods). The resulting structural map shows that MD and TrajCast occupy the same region of the τ – Q plane, departing from equilibrium at nearly identical points, confirming that TrajCast faithfully captures the structural evolution observed in MD.

Despite training only on liquid water at ambient conditions, TrajCast reproduces the glass transition and matches experimental observations

on HGW, extrapolating to both unseen configurations and velocities far outside its training regime. A recent study demonstrated that MLIPs can also reproduce glassy water without explicit training on amorphous structures⁶⁸; however, their training set required the inclusion of crystalline ice configurations. In contrast, TrajCast achieves comparable accuracy with a far smaller and less diverse training set, highlighting its data efficiency and generalizability.

Scalability and data efficiency. So far, we have presented results for a single prediction horizon and fixed training set size. To explore their impact, we vary both for all three systems (Fig. 6), training three models per setting with different initial weights. To facilitate this comparison, we express prediction horizons as multiples of the MD time step (MTM) and focus on the relative velocity MAE.

We first vary Δt and keep the number of training samples identical to the previous sections. Figure 6a shows parity plots for water at MTMs of 1 and 30, with relative velocity errors of −0.03% and −4.64%. Despite visually appearing well-performing, none of the MTM-30 models produced stable trajectories beyond a few tens of thousands of steps. While MLIPs tolerate relative force errors up to 30%⁶⁹, TrajCast requires substantially smaller errors for stable roll-outs. Figure 6b shows the relative velocity MAE across MTMs of 1, 2, 10, 20 and 30 for all the systems, with water exhibiting the steepest increase. The maximum acceptable error is system-dependent: quartz remains stable at a 3.96% relative error at 30 MTM, whereas the water model at an MTM of 20 and a relative velocity MAE of 1.62% may already produce instabilities. MAEs for additional MTMs and the scaling of the displacement

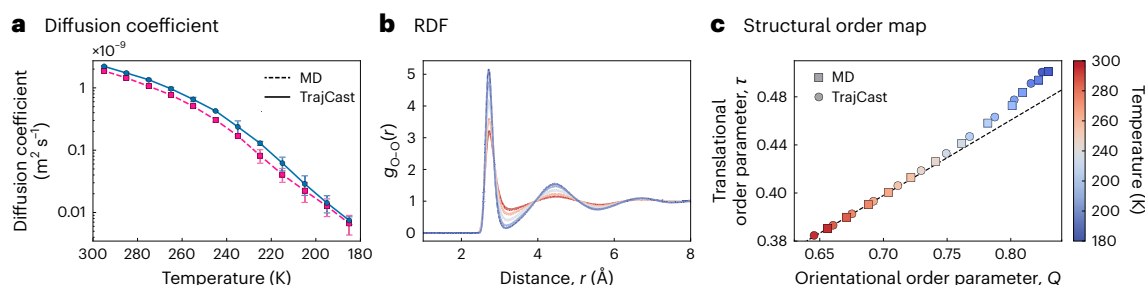


Fig. 5 | Zero-shot prediction of the amorphization of liquid water. a, The evolution of the diffusion coefficient upon cooling from 300 to 180 K, computed from MD and TrajCast *NVT* trajectories. The points are mean values across four independent runs, with error bars corresponding to the standard deviation. The line serves as a guide to the eye. **b,** Changes in the water structure upon cooling, as captured by selected oxygen–oxygen RDFs from MD and TrajCast at varying temperature, showing almost identical behaviour. Each RDF is coloured according to its respective temperature. **c,** The relationship between

the translational (τ) and orientational (Q) order parameters during cooling. MD and TrajCast follow the same structural evolution, which is initially linear in equilibrium, and depart from this trend at nearly identical points. The dashed line marks the linear equilibrium fit to the MD data between 295 and 265 K. Markers are coloured according to the respective temperature. All properties are averaged over 1-ns segments, corresponding to 10-K temperature intervals, with the resulting value assigned to the midpoint temperature of each segment.

MAE, following a similar trend albeit with consistently smaller relative errors by about 40%, are shown in Supplementary Sections A.1 and A.2.

An intuitive solution to stabilize large MTM models is to train on larger datasets. To assess the data efficiency, we train models on 10% and 50% of the original dataset sizes. The resulting learning curves follow a power law⁷⁰, where the exponent s determines how quickly the performance improves with additional data. Figure 6c shows s as a function of the MTM for all systems, with the schematic inset illustrating that s corresponds to the slope of a straight line in a log–log plot (for full learning curves, see Supplementary Section A.2). Paracetamol consistently shows the largest $|s|$, followed by water and quartz, reflecting the increasing constraints on atomic motion as one moved from isolated intramolecular dynamics to correlated molecular motion to lattice-coupled vibrations. Notably, these scaling exponents are up to four times larger than those reported for MLIPs on similar systems^{26,28}, suggesting that incorporating velocities extracts more information from each configuration. This enables the training of accurate TrajCast models on just a few hundred picoseconds of trajectory data, compared with the hundreds of nanoseconds required by other methods^{44,45,47,50}. This is crucial for scaling to ab initio quality datasets across multiple systems and building foundation models.

Conclusion

We introduce TrajCast, a methodology based on autoregressive equivariant MPNNs to accelerate the generation of accurate MD trajectories. Instead of predicting forces for numerical integration, our framework directly updates velocities and positions, bypassing the need for small integration steps. We have benchmarked our methodology on a variety of chemical systems ranging from a small molecule to equilibrium condensed matter phases, such as a crystalline material and a bulk liquid, to out-of-equilibrium and kinetically arrested dynamics. In all cases, TrajCast reliably reproduces structural, dynamical and energetic properties, while enabling $10\times$ to $30\times$ larger time steps than traditional, atomistic MD simulations. We also demonstrate that it can explore regions of phase space absent from the training data, generating physically meaningful ensembles across metastable states and escaping local free energy minima. Importantly, TrajCast is transferable and scales well to larger systems, allowing the generation of over 1 ns of trajectory data per day for 1,728 water molecules and more than 15 ns for a quartz supercell with more than 4,300 atoms. In contrast, the current MACE MLIP architecture enables MD simulations of about 1 ns for 1,000 atoms per day³⁶. Notably, our models are trained using only a few hundred picoseconds of trajectory data, a quantity feasible to generate with ab initio MD. Despite constraining the training set in size and thermodynamic conditions, TrajCast demonstrates impressive

zero-shot generalization, accurately reproducing the amorphization of water upon cooling to 180 K using a model trained exclusively on liquid water under ambient conditions. Far beyond its training distribution, it captures both the structural and dynamical changes associated with the glass transition, in quantitative agreement with MD.

The current version of TrajCast operates with a fixed forecast horizon, but future work will explore multiple time-step training, which could improve the generalizability and accuracy over longer timescales, as demonstrated by a recent autoregressive GNN for weather forecasting⁷¹. The current force-free approach of TrajCast prevents the computation of pressure and other force-related properties, restricting forecasting to the microcanonical (*NVE*) and canonical (*NVT*) ensembles. Extending the methodology to the isobaric–isothermal (*NpT*, where p is pressure) ensemble, however, will be an interesting avenue for future work and may be achievable by coupling TrajCast to classical or machine-learning force fields for pressure control, as suggested by recent work⁷². Additionally, although we benchmarked our methodology on various systems beyond isolated molecules, it would be interesting to explore how it performs when trained on multiple, chemically diverse materials or molecules simultaneously. Similar to recent universal MLIPs, which yield reasonable predictions for hydrogen combustion and nanoconfined water despite being trained on inorganic crystals³⁶, this could mark an important step towards a foundation model for MD capable of interpolating across chemical space. While all reference data in this work were generated using classical force fields, TrajCast can, in principle, be trained on trajectories from first-principles many-body potential energy surface. Importantly, this does not necessarily require larger datasets, as MLIPs have been shown to reach the coupled cluster with single, double, and perturbative triple excitations (CCSD(T))-level accuracy without needing more data than for density functional theory²⁶. Further, recent work on meta-learning suggests that cheaper or approximate data can be used to pretrain models, thereby reducing the amount of costly reference data required⁷³.

While TrajCast has the potential to accelerate atomistic simulations and material discovery in general, our zero-shot experiments on hyperquenched glassy water highlight its particular promise for exploring slow physical phenomena at low temperatures. In these systems, slow relaxation times require long simulation runs, as seen in a recent study⁷⁴ on supercooled water that took several years to reach the microsecond scale, and impose higher correlations, potentially enabling one to leverage larger forecasting horizons. Thus, TrajCast could greatly accelerate this process, making large-scale simulations more efficient. By addressing the long-standing challenge of accessing long simulation times, our work could help bridge currently disjoint length scales, which is a key goal in many scientific disciplines.

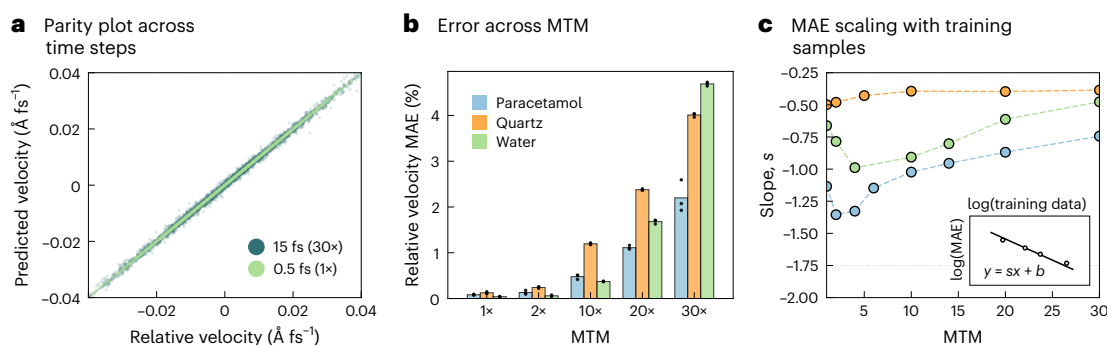


Fig. 6 | The impact of the prediction horizon and training set size on the performance of TrajCast. **a**, A parity plot comparing the predicted and reference velocities for TrajCast models trained for water with a prediction horizon of 0.5 and 15 fs, respectively. **b**, The dependence of the relative MAE velocity on the MTM. Bar heights show the mean across three models with different initial weights, represented by the markers. The relative velocity error is the ratio of the velocity MAE and the mean absolute velocity. **c**, The exponents from power-law

fits to learning curves, quantifying the relationship between the model accuracy and training set size across different chemical systems and MTM. The inset shows that these exponents correspond to the slope s of a linear fit on a log–log plot of MAE versus training set size. The larger the absolute value of s , the larger the improvement on the generalization error of the model when additional data are used. The dashed lines serve as guides to the eye.

Methods

Equivariant message-passing neural networks for atomistic systems

MPNNs, a subclass of GNNs, learn mappings from molecular graphs to target properties such as the potential energy in MLIPs. In these graphs, nodes represent atoms, connected by edges usually determined through a distance criterion. Each node i has attributes such as its atomic type Z_i and position $\mathbf{r}_i$ in three-dimensional (3D) Euclidean space. Depending on the task, additional attributes may include the velocity $\mathbf{v}_i$ or the force $\mathbf{f}_i$. Edge attributes typically consist of the Euclidean vectors $\mathbf{r}_{ij}$ connecting two nodes i and j .

The local state of each node is encoded in latent features $\mathbf{h}_i^t$ that are iteratively updated over T message-passing layers. In each layer t , nodes exchange information via messages $\mathbf{m}_{ij}^t$, aggregated from all neighbours and used to update the node's latent state. Formally, this is expressed as⁷⁵

$$\mathbf{m}_i^t = \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^t = \sum_{j \in \mathcal{N}(i)} M^t(\mathbf{h}_i^t, \mathbf{h}_j^t, \mathbf{e}_{ij}) \quad (1)$$

$$\mathbf{h}_i^{t+1} = U^t(\mathbf{h}_i^t, \mathbf{m}_i^t) \quad (2)$$

where $\mathcal{N}(i)$ represents the set of nodes connected to node i and M^t and U^t are arbitrary message and update functions for layer t . While the original formulation omits node attributes in the update, advanced MLIPs such as BOTNet²⁷ or MACE^{28,76} and other MPNN architectures such as SEGNNs⁷⁷ incorporate them to prevent information loss across layers. After T message-passing steps, the final features may be pooled for global property prediction or passed through a readout function directly for node properties.

The latent features $\mathbf{h}_i^t$ can consist of various types of geometric objects, such as scalars, vectors or higher order tensors, each transforming uniquely the Euclidean group $E(3)$ operations of translation, rotation and inversion. In practice, however, it suffices to focus on the orthogonal group $O(3)$ (rotations and reflections), as translational invariance is ensured by using interatomic vectors rather than absolute positions. Transformations under $O(3)$ are described by irreducible representation (irreps) that are labelled by their rotation order $l = 0, 1, 2, \dots$ and parity $p \in \{1, -1\}$. Scalars ($l = 0, p = 1$) remain invariant under rotation, while vectors ($l = 1, p = -1$) transform consistently (equivariantly) with the geometry. If the features only depend on the atomic positions, formally each element of the feature vector with rotation order L , $\mathbf{h}_{i,L}^t$, is equivariant under any rotation $\mathbf{g} \in O(3)$ if

$$\mathbf{h}_{i,L}^t(R \cdot (\mathbf{r}_1, \dots, \mathbf{r}_N)) = D_L(\mathbf{g}) \mathbf{h}_{i,L}^t(\mathbf{r}_1, \dots, \mathbf{r}_N) \forall \mathbf{g} \in O(3) \quad (3)$$

where R is the 3D rotation matrix parametrized by $\mathbf{g}$ and $D_L(\mathbf{g})$ is the Wigner D -matrix representing the equivalent transformation in feature space. These symmetries can, in principle, be learned from augmented datasets comprising rotated configurations. Alternatively, they can be explicitly incorporated into MPNNs by using equivariant message and update functions, ensuring that the network respects rotational symmetry by construction. The latter has become common practice in state-of-the-art MLIPs^{26–29}, where leveraging higher order features beyond scalars has been shown to enhance accuracy.

Ensembles and thermostats

By numerically integrating the equations of motion, MD simulations produce a trajectory that follows the distribution of the microcanonical (NVE) ensemble, with a constant number of particles N , volume V and total energy E . In this ensemble, both the total linear momentum $\mathbf{P}$ and the angular momentum $\mathbf{L}$ are conserved. However, since experiments are usually conducted at constant temperature rather than energy, MD simulations are coupled to a heat bath via a thermostat to sample from the canonical (NVT) ensemble, where the temperature T fluctuates around the desired value. Microscopically, the temperature T of an N -particle system is defined by

$$T = 2 \frac{K}{N_f k_B} \quad \text{with } K = \frac{1}{2} \sum_{i=1}^N m_i |\mathbf{v}_i|^2 \quad (4)$$

where K is the kinetic energy, m_i is the mass of particle i , k_B is the Boltzmann constant and N_f corresponds to the degrees of freedom, typically given as $3N - 3$ for a 3D periodic system.

Velocity rescaling is the simplest method for adjusting a system to a target temperature, where atomic velocities are rescaled at a pre-defined frequency by a factor $\alpha = \sqrt{\hat{K}/K}$, with $\hat{K}$ representing the target kinetic energy. However, this method has several drawbacks, including the failure to sample from the canonical ensemble, the generation of discontinuities and the violation of equipartition⁷⁸. The CSVR thermostat⁵¹ overcomes these limitations by sampling $\hat{K}$ from a canonical equilibrium distribution and rescaling the velocities after each integration step. The coupling strength, defined by the relaxation time τ relative to the integration time step, must be chosen appropriately to maintain the target temperature without substantially altering the system's dynamics.

TrajCast methodology

Compared with MLIPs, our model takes the atomic velocities $\mathbf{v}(t)$ in addition to the positions $\mathbf{r}(t)$ and atomic types Z as input. In an embedding layer (Fig. 1b), both node and edge attributes are encoded to

produce the initial node features $\mathbf{h}_i^0$ and express the edge and velocity vectors in a radial and spherical harmonic basis. For chemical element Z_i and a magnitude of the edge vectors r_{ij} , we follow the same approach as the MLPs NequIP²⁶ and MACE²⁸, using a one-hot encoding and learnable Bessel functions multiplied by a polynomial cut-off, as introduced originally in DimeNet⁷⁹, respectively. Since the atomic velocity depends on the particle mass, we encode its magnitude v_i using a Gaussian basis set, where each Gaussian is weighted according to the chemical element of node i . Thus, the μ th element of the encoding is given by

$$R_\mu(v_i) = \sum_{m=1}^M W_{Z_i,m} \phi_m(v_i) \quad (5)$$

where ϕ_m is the m th Gaussian and $W_{Z_i,m}$ is a learnable weight dependent on the atomic species Z_i .

The latent features are updated through a series of convolutions in the message-passing layer, highlighted by the white box in Fig. 1a. The message from node j to node i in layer k , schematically shown in Fig. 1e, is constructed similarly to that in NequIP²⁶ and given by

$$\mathbf{m}_{ij}^k = \mathbf{h}_j^k \otimes_{\text{CG}}^{\text{MLP}_{\text{CG}}^k(R(r_{ij}))} \mathbf{Y}(\hat{\mathbf{r}}_{ij}). \quad (6)$$

Here, $\otimes_{\text{CG}}^{\text{MLP}_{\text{CG}}^k(R(r_{ij}))}$ represents the Clebsch–Gordan tensor product, which generates all equivariant combinations of irreducible representations (irreps) of node features and the spherical harmonic embedding of the edge vector projected on the unit sphere, $\hat{\mathbf{r}}_{ij}$. The individual terms are then weighted by a multilayer perceptron (MLP), which takes the radial basis embeddings of the edge vector magnitudes r_{ij} as input. For details on the derivation, the reader is referred to textbooks on representation theory overview and refs. 77,80 or²⁷ for applications to GNNs and atomistic modelling.

The constructed pairwise messages, $\mathbf{m}_{ij}$, only contain information about the topology of the neighbourhood $\mathcal{N}(i)$. To incorporate velocities in the feature update (Fig. 1d), we follow an approach introduced by Brandstetter et al.⁷⁷. After an average pooling of the pairwise messages $\mathbf{m}_{ij}$, we compute a node-based Clebsch–Gordan tensor product between the aggregated message, $\mathbf{m}_i$, and the spherical harmonic embedding of the velocity vector projected on the unit sphere, $\hat{\mathbf{v}}_i$, resulting in

$$\tilde{\mathbf{m}}_i^k = \mathbf{m}_i^k \otimes_{\text{CG}}^{\text{MLP}_{\text{CG}}^k(R(v_i, Z_i))} \mathbf{Y}(\hat{\mathbf{v}}_i). \quad (7)$$

Similar to the first tensor product, the respective paths are weighted on the basis of the species-dependent embedding of the velocity magnitudes, $R(v_i, Z_i)$, passed through an MLP. We constrain the dimensionality of $\tilde{\mathbf{m}}_i^k$ by applying a linear self-interaction after the average pooling step. The new node features $\mathbf{h}_i^{k+1}$ are then computed in a ResNet-like approach: The conditioned message $\tilde{\mathbf{m}}_i^k$ is passed through a nonlinear gate⁸¹ and added to the transformed node features from the previous layer, which are processed using an element-dependent linear self-interaction layer, as proposed in MACE^{28,76}.

After K message-passing layers, the vector components ($l = 1, p = -1$) of the final latent features $\mathbf{h}_i^K$ are passed through two linear layers to produce the initial estimate of the displacement vector $\Delta \mathbf{r}_i'$ and the new velocity $\mathbf{v}_i'$. Predicting relative displacements instead of absolute positions makes the model predictions independent of the dimensions of the simulation box and preserves translational invariance. While these predictions may already be close to the reference dynamics, consistency with the microcanonical ensemble requires that the total linear and angular momentum be conserved. We enforce this directly within the architecture by removing any centre of mass motion, $\Delta \mathbf{D}_{\text{COM}}$, and excess momenta, $\Delta \mathbf{P}$ and $\Delta \mathbf{L}$, from $\Delta \mathbf{r}_i'$ and $\mathbf{v}_i'$ as illustrated in Fig. 1d. Importantly, this refinement is applied during both training and roll-out, so the model weights are optimized with these constraints enforced. The new positions are then obtained by adding the adjusted displacements to the current atomic positions.

While TrajCast can, in principle, generate accurate trajectories in the *NVE* ensemble iteratively by using each output as input for the next step, long autoregressive roll-outs tend to gradually accumulate small prediction errors. Over time, these errors can lead to drift in positions and velocities, potentially causing instability and unphysical behaviour. To address this and produce trajectories consistent with the canonical ensemble, we apply the CSVR thermostat⁵¹ after each step, rescaling the velocities predicted by the MPNN by a stochastic factor determined by the target temperature. Noise injection has previously been shown to help mitigate error accumulation in turbulence modelling⁸². The CSVR thermostat further provides additional stability by maintaining a physically consistent distribution of atomic velocities during long roll-outs. The resulting positions $\mathbf{r}_\Delta$ and velocities $\mathbf{v}_\Delta$, moreover, follow the Boltzmann distribution, allowing for comparison with experiments at constant temperature. After a first version of this manuscript appeared online, Bigi et al.⁷² introduced FlashMD, which builds on ideas similar to TrajCast, albeit with a different architecture and implementation. While FlashMD demonstrates long time-step forecasts for various systems, in contrast to TrajCast, its reliance on local thermostats and the associated perturbation of the dynamics constrains validation primarily to equilibrium properties.

We train our equivariant MPNN on configurations generated during MD runs within the *NVE* ensemble using common time steps of 0.5–1.0 fs. Relative displacements and new velocities are then computed over larger time intervals Δt . Displacements are calculated as the difference between the current and next configurations, accounting for periodic boundary conditions, while the new velocities correspond to those at $t + \Delta t$. During training, the model weights are adjusted to minimize the residual between the true and predicted displacement and velocity vectors. Importantly, this training procedure is independent of any thermostat, ensuring that the MPNN learns the deterministic *NVE* dynamics. During inference, predictions can then be lifted to the canonical ensemble via the CSVR thermostat. Since the current version of TrajCast infers the prediction horizon directly from the data, the training set needs to be precomputed at a fixed Δt . Similar to traditional MD simulations, the choice of Δt involves a trade-off: Smaller values inherently result in higher accuracy, as the system becomes increasingly decorrelated over time, but are less efficient, requiring more forward passes to produce the desired trajectory. By predicting atomic displacements and velocities on the basis of the local environment encoded in latent features, our approach ensures scalability to much larger systems than those encountered during training. Unlike MLIP-driven MD, TrajCast operates in a force-free manner and does not rely on computing gradients with respect to the positions to propagate the system in time. Our framework builds on the e3nn⁸³ and cuEquivariance Python libraries, which ensure the efficient equivariant transformation and propagation of geometric objects within deep learning frameworks.

Reference data sets

All data used for training and validation were generated using classical MD simulations with the LAMMPS^{84,85} software package. Training trajectories were generated in the *NVE* ensemble, while data to assess the roll-out performance of the model architecture stem from *NVT* runs. The *NVE* samples are labelled with atomic displacements to the next state at $t + \Delta t$, accounting for periodic boundary conditions where applicable, and the corresponding velocities, which serve as training targets for the MPNN. To ensure that the *NVE* trajectories are at the desired temperature, we first equilibrate the system in the *NVT* ensemble. During the equilibration period, the total energy is sampled, and configurations with a total energy close to the distribution's average are subsequently selected. The *NVE* runs are then initiated from five configurations that are uniformly sampled from this subset, ensuring that the selected snapshots are at least 1 ps apart. The specific simulation settings for each system are detailed below.

Paracetamol. An isolated paracetamol molecule is simulated in vacuum using the OPLS all-atom force field^{86,87} instantiated with Moltemplate⁸⁸. We use a time step of 0.5 fs owing to the presence of hydrogen atoms. The system is first equilibrated in the *NVT* ensemble for 25 ps (50,000 steps) using a Nosé–Hoover chain thermostat⁸⁹, with ten chains, a damping factor of 5 fs and a target temperature of 300 K. This equilibration protocol is applied uniformly before subsequent production runs in both the *NVE* and *NVT* ensembles. The *NVE* production runs are initiated from configurations where the total energy is within a relative threshold of 0.001 from the mean of the distribution. For the *NVT* production run, we utilize the CSVr thermostat⁵¹, consistent with the set-up in TrajCast, with a relaxation time of 50 fs. The temperature is calculated on the basis of $3N - 6$ degrees of freedom. Net angular momentum is removed at each time step during *NVT* simulations, while in the *NVE* ensemble, this quantity is naturally conserved, maintaining an initial value of 0. Production simulations are conducted for 100 ps (200,000 steps) in the *NVE* ensemble to generate training data, and for 10 ns (20,000,000 steps) in the *NVT* ensemble for validation.

Quartz. Bulk α -quartz containing 162 atoms is simulated using the van Beest–Kramer–Santen potential⁹⁰ with truncated Coulomb interactions⁹¹ and a time step of 1 fs. The system is equilibrated to 300 K in two stages: first in the *NpT* ensemble at ambient pressure using the Berendsen barostat⁹² for 10 ps (10,000 steps) with a relaxation time of 0.1 ps, then in the *NVT* ensemble for 50 ps (50,000 steps) with a Nosé–Hoover chain thermostat⁸⁹ with three chains and a relaxation time of 0.1 ps. Production runs of 100 ps (100,000 steps) in the *NVE* ensemble are initiated from five configurations randomly sampled from the equilibrated *NVT* trajectory, selecting those within a ± 10 K range of the target temperature. The minimum time separation between selected configurations is set to 0.1 ps. *NVT* simulations for comparison with TrajCast predictions are carried out using the CSVr thermostat⁵¹ with a relaxation time of 100 fs and are performed for 15 ns (15,000,000 steps).

Liquid water. Bulk water is modelled using a system of 64 water molecules in a cubic simulation box with a fixed length of 12.4172 Å. Intramolecular and intermolecular interactions of water are described by the flexible version of the simple point charge interaction potential^{93–95}. All MD simulations are carried out with a time step of 0.5 fs. The system is equilibrated for 10 ps (20,000 steps) in the *NVT* ensemble using a Nosé–Hoover thermostat⁸⁹ with a relaxation time of 0.5 ps and a target temperature of 300 K. The temperature is calculated on the basis of $3N - 3$ degrees of freedom. Production runs of 50 ps (100,000 steps) in the *NVE* ensemble are initiated from five configurations randomly sampled from the equilibrated trajectory, selecting those within a ± 20 K range of the target temperature. The minimum time separation between selected configurations is set to 0.1 ps. Simulations in the *NVT* ensemble are carried out using the CSVr thermostat⁵¹ with a relaxation time of 50 fs and are performed for 2.5 ns (5,000,000 steps).

Hyperquenched glassy water. For the zero-shot experiments on hyperquenched water, we increase the system size from 64 to 216 water molecules. The density is kept the same as in the liquid water simulations at ambient conditions, and the potential energy surface is also described by the flexible simple point charge model^{93–95}. After a short equilibration at 300 K, we gradually reduce the temperature using the CSVr thermostat⁵¹ at a cooling rate of $q = 10$ K ns⁻¹, identical to that used in a recent MLIP study⁶⁸. The time step is set to 0.5 fs, and we use a relaxation time of 50 fs for the thermostat. In total, we perform four independent runs starting from different initial seeds, each with a total length of 12 ns. The translational order parameter τ , used to quantify the long-range translational order of the system^{96,97}, is defined as

$$\tau = \frac{1}{\zeta_c} \int_0^{\zeta_c} |g_{O-O}(\zeta) - 1| d\zeta \quad (8)$$

where $\zeta = r\rho_n^{1/3}$ is the distance r between the oxygen atoms of pairs of water molecules scaled by the mean nearest-neighbour separation at the number density $\rho_n \equiv N/V$. Further, $g_{O-O}(r)$ is the oxygen–oxygen RDF, which is integrated up to the cut-off $\zeta_c = 2.577$ corresponding to 8 Å (ref. 65). The tetrahedral order parameter Q , conversely, measures the short-range orientational structure and is given by

$$Q = 1 - \frac{3}{8} \sum_{j=1}^3 \sum_{k=j+1}^4 \left(\cos \psi_{jk} + \frac{1}{3} \right)^2 \quad (9)$$

where ψ_{jk} is the angle formed by the lines joining the oxygen atoms of a given molecule and those of nearest neighbours j and k (≤ 4). Following ref. 65, we limit to the four oxygen atoms that are closest to a given oxygen atom. Therefore, this definition of the tetrahedral order parameter is slightly different from the original⁹⁷.

Model size and hyperparameters

All equivariant MPNNs in this work use both even ($p=1$) and odd ($p=-1$) parity features up to a rotation order of $l_{\max} = 2$, creating a feature vector of scalars, vectors and higher-order tensors. We use 64 channels per object, resulting in a feature vector described as a direct sum of irreps and expressed in e3nn⁸³ notation as $64 \times 0o + 64 \times 0e + 64 \times 1o + 64 \times 1e + 64 \times 2o + 64 \times 2e$, where each term specifies the multiplicity (first number), rotation order l and parity p ('o' for odd, 'e' for even). For all experiments, we employ four message-passing layers and use the Sigmoid Linear Unit and tanh as activation functions in the nonlinear gates⁸¹ for even and odd invariants, respectively.

The TrajCast MPNN contains approximately 2.2 million learnable weights, most of which reside in the MLPs that process the radial embeddings of edge and velocity magnitudes. Each MLP comprises 3 hidden layers with 64 neurons per layer and uses Sigmoid Linear Unit activation functions. Identical to NequIP²⁶, for the radial embedding of the edge lengths, we use eight learnable Bessel functions and a polynomial cut-off with $p=6$ (ref. 79). Similarly, eight Gaussian basis functions are used to generate the encoding of atomic velocity magnitudes. The element-specific velocity encoding from equation (5) is concatenated with the chemical species embeddings of each node to form the initial features passed to the first message-passing layer.

The cut-off radius for defining edges between nodes varies by system and is set to 4.0 Å, 4.5 Å and 6.0 Å for paracetamol, quartz and liquid water, respectively. The upper bound of the radial embedding of the velocities is determined by the lightest element, $Z_{\min}$ (hydrogen for water and paracetamol but oxygen for quartz), which governs the maximum velocity. For this element, we calculate the theoretical expectation value of the r.m.s. velocity, $v_{\text{RMS},Z_{\min}}^{\text{MB}}$, on the basis of the Maxwell–Boltzmann distribution. When training across multiple temperatures, we recommend computing the r.m.s. velocity for the highest temperature. We found that the upper limit of the encoding works then best when set to values between $3.5 \times v_{\text{RMS},Z_{\min}}^{\text{MB}}$ and $5 \times v_{\text{RMS},Z_{\min}}^{\text{MB}}$.

In all models, we refine the velocities predicted by the readout to remove any excess total linear momentum. While angular momentum can be easily removed for isolated molecules such as paracetamol, a proper treatment is needed for periodic systems such as bulk quartz or liquid water, where particles crossing boundaries and interacting with periodic images can introduce discontinuities and torques. A common approach is to treat the simulation box as an open system, as suggested by Kuzkin⁹⁸. However, for the bulk systems investigated here, angular momentum conservation is typically not a relevant concern compared with more complex systems such as interfaces or multiphase systems, owing to their isotropic nature and symmetric forces. In fact, we find that all MPNNs perform well even without additional constraints, with no noteworthy excess angular momentum generated during the roll-out.

Training

All models are trained on the basis of the loss function

$$\mathcal{L} = \frac{1}{3 \sum_{b=1}^{N_b} N_a} \sum_{b=1}^{N_b} \sum_{i=1}^{N_a} \lambda_d (\widehat{\Delta r}_{i,v} - \Delta r_{i,v})^2 + \lambda_v (\widehat{v}_{i,v} - v_{i,v})^2 \quad (10)$$

where N_b denotes the batch size, N_a is the number of atoms in each configuration within the batch, $\widehat{\Delta r}_{i,v}$ and $\widehat{v}_{i,v}$ represent the predicted v component of the displacement and velocity of atom i , respectively, and λ_d and λ_v control the relative weighting of the displacement and velocity mean squared error (MSE). For all experiments and independent of the chosen prediction horizon of the model, we assign equal weights $\lambda_d = \lambda_v = 0.5$. The target vectors are normalized by their r.m.s. computed over the entire training set, and the model's predictions are rescaled to match the correct magnitude.

The weights are optimized with the Adam optimizer⁹⁹ using AMS-Grad¹⁰⁰ and default values for $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The number of training samples varies across experiments, with models trained on 1,000, 5,000 and 10,000 configurations of paracetamol and 500, 2,500 and 5,000 snapshots of quartz and liquid water. The results reported in the experiments correspond to models trained on the maximum sample size, unless stated otherwise. All training configurations are drawn randomly from three independent *NVE* trajectories of each system. The validation set comprises a quarter of the training set size, and a fixed hold-out test set of 1,000 configurations is used to compute the final accuracy metrics. Both the validation and test set are sampled from independent trajectories, ensuring no overlap with the training set.

Irrespective of the training set size, a batch size of ten for molecules and two for the condensed matter systems is used. The initial learning rate of 0.01 is adjusted by an on-plateau scheduler that reduced the rate by a factor of 0.8 if the validation loss does not improve over 20 epochs. Training continues for a maximum of 1,500 epochs, with early stopping if no validation loss reduction is observed over 100 epochs. Hyperparameters for the optimizer, learning rate and batch size are optimized for the best balance between training time and model performance on the largest training set and prediction horizons. Every experiment is performed using double precision. Both the model and training routine are implemented using the PyTorch library¹⁰¹. All models are trained on a single NVIDIA V100-SXM2-32GB graphics processing unit, with training durations of up to 5, 10 and 11 days for paracetamol, quartz and water, respectively, many completing earlier owing to early stopping.

Trajectory forecasting

All trajectories generated using TrajCast are produced autoregressively by rolling out the predictions of the MPNN, with velocities being rescaled after each step using a CSVR thermostat⁵¹. As in classical MD simulations, the relaxation time of the thermostat, τ , plays a crucial role and must be carefully chosen. In Supplementary Section B.1, we explore how TrajCast behaves with different values of τ for all systems and varying Δt . We find that, for larger prediction horizons, much lower relaxation times are required compared with typical MD values to maintain the target temperature and fluctuations. This is likely due to TrajCast implicitly performing multiple time steps before rescaling, whereas in MD, velocities are rescaled after each step. For the reported experiments, we set τ to $10 \times \Delta t$ (70 fs) for paracetamol, $2.5 \times \Delta t$ (75 fs) for α -quartz and $10 \times \Delta t$ (50 fs) for liquid water, both at constant temperature and under cooling. The target temperature for all systems is 300 K and is defined on the basis of $3N - 6$ and $3N - 3$ degrees of freedom for paracetamol and the condensed matter systems, respectively. In the zero-shot experiment on the amorphization of liquid water, the target temperature is decreased linearly at a constant rate of $q = 10 \text{ K ns}^{-1}$. Initial velocities are sampled from a Gaussian distribution at 300 K with zero net linear and angular momentum. For paracetamol, the net angular momentum is zeroed every 100 forward passes to correct for the excess angular momentum induced by the CSVR thermostat.

For paracetamol, we produce a trajectory of 7 ns (1,000,000 forward passes), while for water and quartz, the roll-out is performed over 500,000 iterations, corresponding to 2 ns and 15 ns, respectively. These systems are equilibrated for 10% of the total number of forward passes used for production. The cooling of liquid water to hyperquenched glassy water is carried out over 12 ns, equivalent to 2,400,000 autoregressive steps and representing our longest roll-out. In addition to PyTorch¹⁰¹, e3nn⁸³ and cuEquivariance, we also use the ASE¹⁰² python package to efficiently generate and save the trajectories.

Data availability

The model weights used in the experiments on various chemical systems are available via Hugging Face at <https://huggingface.co/ibm-research/trajcast.models-arxiv2025> (ref. 103). The training, validation and testing datasets for these experiments are available via Hugging Face at <https://huggingface.co/datasets/ibm-research/trajcast.datasets-arxiv2025> (ref. 104). Source data are provided with this paper.

Code availability

An open-source implementation of TrajCast for training models and performing trajectory forecasting is available via GitHub at <https://github.com/IBM/trajcast> and via Zenodo at <https://doi.org/10.5281/zenodo.19055062> (ref. 105).

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F.L.T. designed research. F.L.T., T.R. and M.E. performed experiments. F.L.T., T.R. and M.E. wrote the code. F.L.T., J.D.O.-P., T.T. and F.M. generated data. All authors analysed data and wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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